From a Boolean Function to an HD Standard Cell

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Abstract

We built a generator that turns a Boolean function of up to four inputs into a static CMOS standard cell in the SKY130 high-density frame. The flow synthesizes and orders the transistor networks, routes the layout, and produces the physical, electrical, and logic views needed to verify the cell and use it in synthesis. All 33 generated cells passed layout-versus-schematic, extracted-circuit, and model checks, and Magic reported zero design-rule violations across 792 tapped abutment contexts. Seven of the nine cells in the detailed layout comparison match the installed cell width. Calibration on six installed cells gave a median timing-table error of 3.36% against the shipped Liberty data. For NOR3, stack sizing reduced the worst rising delay from 330 to 223 ps, while increasing cell area by a factor of 2.5 and roughly doubling input capacitance.

Use of AI. We used AI as a development aid. The generator makes its synthesis, placement, and routing decisions with deterministic algorithms, and the project records AI use in `ai/use_log.md` and `ai/mistakes_log.md`.

1 Introduction

A truth table describes what a cell should compute, but a cell intended for a standard-cell library must also fit next to its neighbors, expose its pins on usable layers, and provide timing data that synthesis and timing tools can read. We built a flow that takes an expression, a truth table, or both in a JSON request and produces a cell in the SKY130 high-density (HD) frame. Every layout is $2.72\,\mu$m tall, and its width is an integer multiple of the $0.46\,\mu$m site width.

Each successful request produces a schematic netlist, GDS layout, extracted SPICE netlists, a Verilog model, a characterized Liberty view, and a JSON response. The response records the generated files and the outcome of the checks, while the combined Liberty file makes the cells available to the technology mapper.

2 Method

2.1 From a request to transistor networks

The request reader enumerates the expression's truth table, using the first listed input as the least significant bit of the row index. When a table is supplied with the expression, the reader checks that they agree. For reference cells, the reader finds a function-preserving permutation from the requested pins to the installed pin names. The truth table follows the same permutation so that every later check uses the same input order.

The synthesizer uses Quine–McCluskey minimization and Petrick's method to find minimum covers, then considers factored forms of the function and its complement. It also tries simple decompositions in which a group of inputs is evaluated as a subfunction and its result feeds another stage. Each stage uses complementary pull-up and pull-down networks with at most three transistors in series. Among the candidates it generates, the synthesizer prefers fewer transistors, followed by fewer inverters and fewer stages. For XOR3, combining a two-input XOR subfunction with the remaining input gives a four-stage, 20-transistor implementation.

The `hd` sizing mode uses the frame's base device sizes and adopts the installed W/L values where devices in a reference cell can be matched. The `stack` mode increases output-stage device width in proportion to the series depth and folds wide devices into parallel fingers. A third mode, `opt`, searches over discrete widths for internal stages using an Elmore delay estimate, with the output stage held at its base sizes. We tested this last mode on six functions to see whether the estimated improvement survived extraction.

2.2 Transistor ordering and geometry

We represent each transistor as an edge between two diffusion nodes, with its gate net as the edge label. A common Euler trail traverses the edges in the pull-up and pull-down graphs in the same gate order. When such an order is available, the two rows can use aligned gate columns and share diffusion between consecutive devices [1]. The ordering code tries legal series permutations, finds covers of the two graphs by paired trails, and considers different stage orders and orientations. Two stages can share diffusion at their interface when the endpoint nets agree in both rows. The layout planner receives several legal orders because a placement with a good diffusion arrangement may still leave too little room for wiring.

For adjacent gates without a contact, the $0.15\,\mu$m gate length and the $0.21\,\mu$m poly spacing rule `poly.2` give a lower bound of $0.36\,\mu$m. With a diffusion contact between the gates, the $0.17\,\mu$m contact and the `licon.11` clearance of $0.05\,\mu$m on either side set the usual pitch:

$$p_{\mathrm{contact}} = (0.15 + 0.17 + 2 \times 0.05)\,\mu\mathrm{m} = 0.42\,\mu\mathrm{m}.$$

In the actual plan, spacing between contact landings or the clearance needed to carry a local interconnect track across the gate band can set a larger pitch. Some crossing channels require 0.68–0.70 μm. At a diffusion break, the two $0.26\,\mu$m end extensions and the $0.27\,\mu$m diffusion spacing

set a gate-to-gate pitch of 0.94 μm. The planner rounds the resulting width up to a whole site. It records the binding constraint for each gap in the layout plan, and the rule-check report records the checks of the 31 rule and frame values used by the planner.

2.3 Routing and physical views

The router first tries Manhattan paths on local interconnect (`li1`) with fixed contact positions. If the fixed positions block a route, a second router can place the signal diffusion contacts where the wires reach the diffusion, subject to the contact and spacing rules. It raises the cost of congested grid points over successive passes, following the negotiated-congestion idea in PathFinder [2]. The planner can also try a different transistor order or a wider layout. Where local interconnect cannot complete the routes, the planner can use first metal (`met1`) for signals and records that decision.

Adjacent gates on the same net can share a poly bar and a contact, which is particularly useful for folded devices. Where spacing allows, the planner adds parallel contact cuts and aligns diffusion contacts with the vias above them. For signal nets that use `met1`, it also reserves room for a via landing at each signal pin so another wire cannot occupy the space needed to reach it. GDSfactory draws the shapes from the layout plan, and Magic extracts the device-only and RC netlists used in the subsequent checks.

3 Verification

3.1 Physical, electrical, and model checks

We verify the cell at several points because each representation can introduce a different error. For example, a correct truth table cannot show whether a device has the wrong width, and a passing device-only LVS does not exercise the RC-extracted circuit during an input transition. Table 1 lists the acceptance tests and the corresponding deliberately faulty controls. The input fixtures and the controls derived by the tests are catalogued in `tests/broken/README.md`.

For each cell, the DRC suite combines three HD neighbors (tap, inverter, and fill) with four placement arrangements and both generated-cell positions, giving 24 tapped contexts. The LVS suite compares the generated layout with its schematic and, for the single-stage HD references, compares the extracted devices with their installed counterparts, including device dimensions. The two multi-stage reference cells are checked against their generated schematics and compared with HD at the function, geometry, and timing levels because their transistor circuits differ.

The extracted-circuit check runs at TT, 1.8 V, and 25 °C, with a 122.5 ps input slew and a 9.52 fF output load. An n-input cell is tested on 2^n static vectors and $n2^n$ directed single-input transitions. Before and after each edge, the output must be within 10% of its expected rail. If an edge leaves the logical output unchanged, the output must also stay on its side of half supply throughout the transition. The test puts independent copies of the circuit into a shared simulation deck, with separate stimuli and loads, so the device models need to be loaded fewer times. Separately, Icarus Verilog exercises the model on every input row, and the Liberty function is evaluated against the same request.

Each ngspice job covers all cases for its cell. The batched check took about 30 seconds per cell, compared with up to 80 separate invocations for a four-input cell, each spending about 45 seconds loading the SKY130 models. The input cases and verdicts were preserved. The combined `make drc lvs compare sim model` run took 32.6 minutes for 33 cells.

3.2 Characterization procedure

The characterizer measures the installed references before characterizing generated cells. It extracts the installed layouts with the same Magic flow used for generated layouts, includes the geometry-dependent source and drain stress distances in the device parameters, and uses the slew and load indices from HD Liberty. Input and output crossings at 50% define propagation delay, while the 20%

Table 1: Verification criteria and deliberately faulty controls.

Check	Acceptance criterion	Fault exercised by the test
Request and rule values	The expression agrees with its truth table, and the layout rule values agree with the PDK or frame.	Supply an inconsistent NAND2 table or reduce the configured poly spacing below `poly.2`.
Switch level	Every input row reaches the correct rail on both the synthesized and reordered networks.	Remove a pull-up from NAND2 or from the output stage of AND2.
DRC and abutment	Magic reports zero violations in each tapped placement context.	Insert a 0.05 μm first-metal stripe into a tapped row.
Layer coverage	The required implant, high-threshold, and poly-contact layers enclose the corresponding geometry.	Remove one of `nsdm`, `psdm`, `hvtp`, and `npc` from a copy of each layout.
LVS and RC-LVS	Netgen matches the extracted devices and schematic, and the collapsed RC netlist preserves connectivity and distinct ports.	Change a NAND2 PMOS width or permute the HD pins of AOI21 so that its function changes.
HD frame and pins	The generated reference cells satisfy the required frame, rail, pin-layer, and pin-position checks.	Remove the inverter's `hvtp` mask or move its input contact out of the poly-contact band.
Transistor simulation	All static vectors and directed single-input transitions meet the rail and settling criteria.	Remove a pull-up from the RC-extracted NAND2 circuit.
Verilog and Liberty	Both models agree with the truth table, and the final Liberty view has the required pins, area, and timing arcs.	Flip a Verilog truth-table row, substitute a wrong Liberty function, or delete an input timing arc.
Characterization	The six installed references meet the timing and capacitance tolerances against shipped Liberty.	Index the timing tables by the full input ramp instead of the threshold-to-threshold slew.
Mapping round trip	Both ABC scripts produce an equivalent mapped netlist.	Swap two inputs in the Liberty function for `f14_1`.

and 80% crossings define slew. For each input arc, the simulator tries the side-input assignments that sensitize it and retains the slowest applicable response. Pins with both timing senses are characterized in both senses. Input capacitance is estimated from the charge drawn during an input transition.

The generated cells use the slew and load indices of the HD NAND2 7×7 template. We extend the settling window at the largest load for cells with weak drive, and record each extension in `extended_windows_ns`. The final Liberty check verifies that the timing tables are complete, that the function and pins match the request, and that the cell area matches the layout. The flow then maps a design for each requested function using the generated cell and the reduced HD library. ABC equivalence compares the resulting circuit with the requested function for each of two mapping scripts.

4 Results

4.1 Verification coverage and cell inventory

The test suite contains 19 `hd` cells, eight `stack` cells, and six `opt` cells. All 33 passed the switch-level checks on both network representations, schematic LVS, RC-LVS, extracted-circuit simulation, and the final Verilog and Liberty checks. The extracted simulations covered 298 static vectors and 1,002 directed input transitions. Magic reported zero violations in all 792 tapped abutment contexts, while the deliberately undersized metal stripe produced one violation. The layer checks passed on the 33 generated cells and two installed controls, and all 132 copies with a required layer removed were rejected. All seven required device-level HD comparisons matched.

We first checked the movable-contact and shared-poly routing changes on `a21oi_1`, `o22ai_1`, `nand3_stack_1`, `o21ai_stack_1`, and `nor4_stack_1` in `build/chk2`. All 120 tapped contexts passed DRC, the layer checks passed, and both LVS and RC-LVS matched, including the installed HD comparison for `a21oi_1`. We also checked the stacked-contact mcon array rule and first-metal pin access on five cells that use signal `met1`. Their checks in `build/pa` covered 120 tapped contexts with zero DRC violations, the layer checks, LVS, RC-LVS, and transistor-level simulation. The updated layouts then passed the full set of checks in the 33-cell suite.

Table 2 reports the realized layout for every request. A finger count can exceed the logical transistor count because wide devices are drawn as parallel devices. A planned break counts one interruption between diffusion segments in the placement. The GDS comparison in Table 3 counts breaks separately in each diffusion row.

4.2 Euler trails and the cost of diffusion breaks

The ordering stage found a one-trail cover for each logic stage in the test suite. Some of those trails cannot be joined in the chosen whole-cell order because the diffusion nets at the interface disagree in at least one row. This accounts for the stage-interface breaks in MAJ3, MAJ4, MUX2, and the folded NOR4. The layout planner can also choose a placement with more breaks than the first Euler candidate when its gate order is easier to route. In folded AOI22, the selected placement has one break although a zero-break ordering exists. The selected XOR3 layout has three breaks and spans 14 sites (Figure 1).

Replacing a $0.42\,\mu$m contacted gate pitch with the $0.94\,\mu$m pitch at a break adds $0.52\,\mu$m before site rounding, which is a little over one site per gap. The final width also depends on routing channels and the position of the pins, so the number of breaks alone does not determine which placement will be narrowest.

4.3 Comparison with installed HD layouts

Figures 2 and 3 show the generated and installed layouts on the same scale within each pair. All nine generated cells pass the frame, signal-pin, and rail checks. INV, NAND2, NOR2, NAND3,

Table 2: Generated cells. Devices are listed as logical transistors followed by drawn fingers. Vias count signal-layer transitions.

Cell	Stages	Devices	Sites	Breaks	Vias
`inv_1`	1	2/2	3	0	0
`nand2_1`	1	4/4	3	0	0
`nor2_1`	1	4/4	3	0	0
`nand3_1`	1	6/6	4	0	0
`nor3_1`	1	6/6	4	0	0
`a21oi_1`	1	6/6	4	0	0
`o21ai_1`	1	6/6	5	0	0
`a22oi_1`	1	8/8	5	0	0
`o22ai_1`	1	8/8	5	0	0
`f14_1`	1	8/8	6	0	0
`f15_1`	1	8/8	5	0	0
`maj4n_1`	1	12/12	10	0	6
`and2_1`	2	6/6	5	0	0
`a21o_1`	2	8/8	6	0	0
`xor2_1`	2	10/10	6	0	6
`maj3_1`	2	12/12	12	1	8
`maj4_1`	2	14/14	10	1	6
`mux2_1`	3	12/12	10	1	0
`xor3_1`	4	20/20	14	3	13
`nand2_stack_1`	1	4/6	5	0	0
`nor2_stack_1`	1	4/6	5	0	0
`nand3_stack_1`	1	6/12	10	0	0
`nor3_stack_1`	1	6/12	10	0	0
`a21oi_stack_1`	1	6/11	7	0	0
`o21ai_stack_1`	1	6/11	8	0	0
`a22oi_stack_1`	1	8/16	10	1	0
`nor4_stack_1`	3	12/14	10	1	0
`and2_opt_1`	2	6/6	5	0	0
`a21o_opt_1`	2	8/8	6	0	0
`xor2_opt_1`	2	10/10	6	0	6
`maj3_opt_1`	2	12/12	12	1	8
`maj4_opt_1`	2	14/14	10	1	6
`mux2_opt_1`	3	12/12	10	1	0

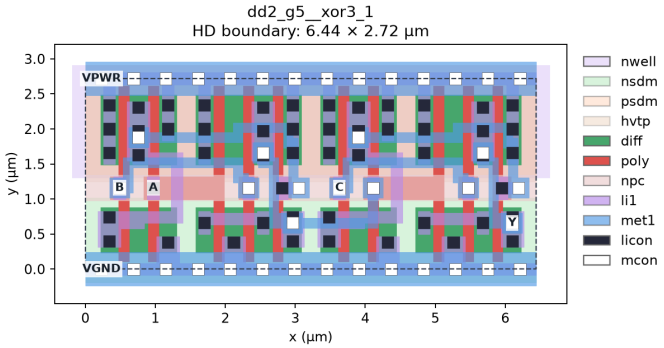


Figure 1: Generated XOR3 layout. The four-stage circuit fits in 14 sites with three planned diffusion breaks and 13 signal vias.

NOR3, AOI21, and AO21 match the installed width. OAI21 and MUX2 each need one additional site, corresponding to 1.2512 μm^2 at the fixed cell height.

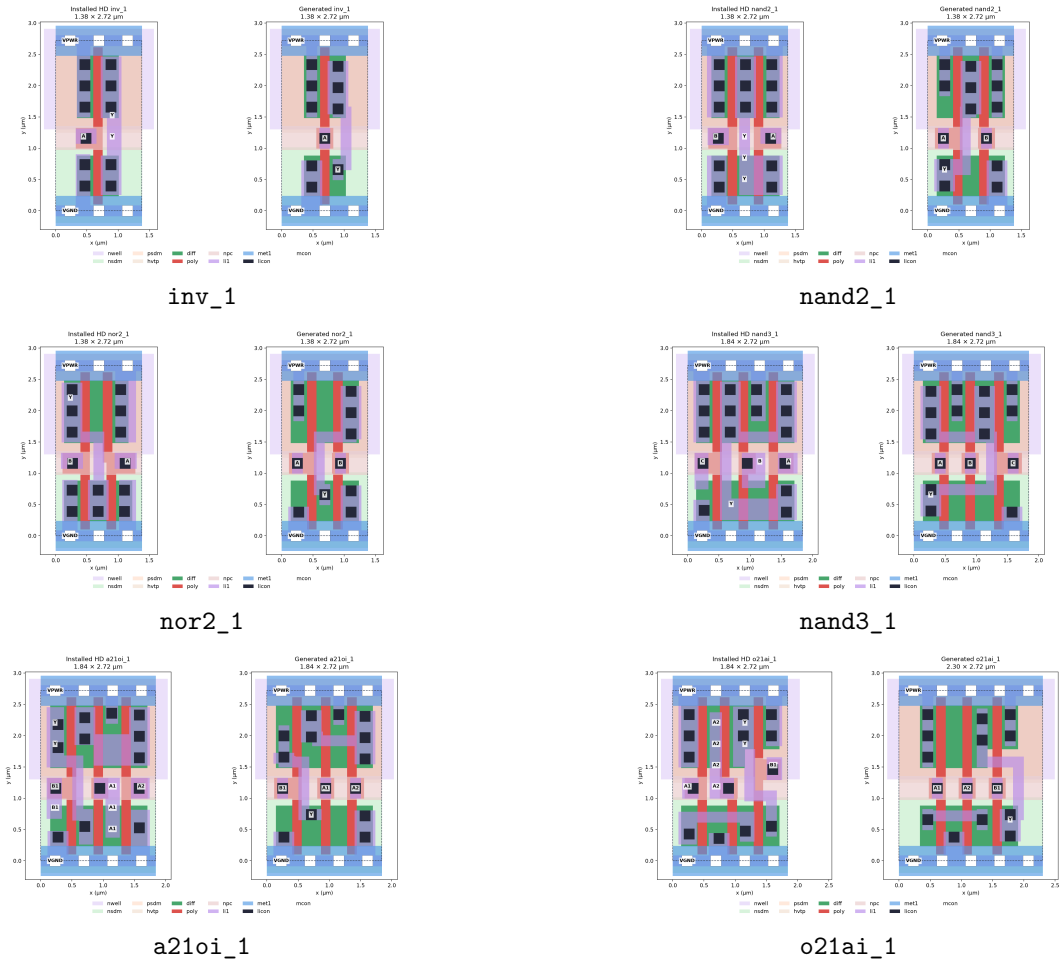


Figure 2: The six required reference cells. Each pair shows the generated cell on the left and the installed HD cell on the right.

Table 3: Layout and extracted output parasitics. Each pair is generated / installed HD. Pitch values follow the physical column order from left to right.

Cell	Sites	Breaks	Gate pitches (μm)	Diff. cuts	$\sum R$ (Ω)	C (fF)
`inv_1`	3/3	0/0	—	9/10	525/392	0.433/0.512
`nand2_1`	3/3	0/0	0.42 / 0.42	12/13	457/459	0.634/0.748
`nor2_1`	3/3	0/0	0.42 / 0.42	9/12	559/1147	0.535/0.657
`nand3_1`	4/4	0/0	0.42, 0.46 / 0.42, 0.48	13/14	806/770	0.984/1.249
`nor3_1`	4/4	0/0	0.46, 0.42 / 0.42, 0.42	9/9	883/915	0.776/1.122
`a21oi_1`	4/4	0/0	0.46, 0.42 / 0.43, 0.445	12/11	643/696	0.451/0.591
`o21ai_1`	5/4	0/0	0.42, 0.48 / 0.48, 0.42	12/11	1044/1600	0.591/0.615
`a21o_1`	6/6	0/1	0.42, 0.48, 0.48 / 0.94, 0.425, 0.43	15/17	610/1751	0.309/0.406
`mux2_1`	10/9	2/0	0.94, 0.46, 0.68, 0.70, 0.68 / 0.545, 0.48, 0.625, 0.48, 0.84	26/16	1037/2095	0.459/0.448

INV has the same single gate column as HD, while NAND2, NOR2, NAND3, and NOR3 reverse the installed left-to-right input order. The two-input gates keep the installed 0.42 μm pitch, whereas the three-input gates have one wider gap to fit their contact and wiring arrangement. The single-stage cells use the installed device dimensions. For most of these devices, the NMOS and PMOS widths

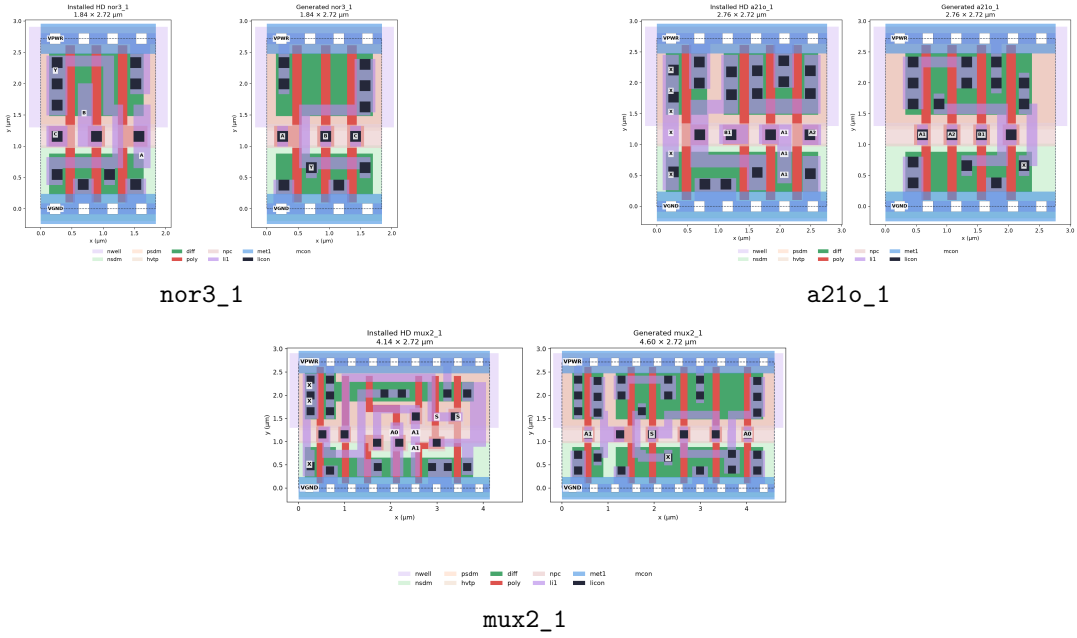


Figure 3: NOR3 and the two multi-stage reference cells. The generated layout is on the left of each pair.

are 0.65 and 1.00 μm, respectively, with a 0.15 μm gate length. The OAI21 B1 PMOS is 0.70 μm wide in both layouts.

AOI21 and OAI21 retain the installed named gate order. The movable-contact router lets AOI21 fit the output and internal routes within four sites on `li1`. The installed OAI21 uses a gate jog and an off-band poly contact that the generator's straight gate columns and band-constrained input contacts cannot reproduce. OAI21 therefore needs five sites even though both diffusion rows are continuous. AO21 and MUX2 have the same transistor counts as their installed counterparts, with eight and twelve devices respectively, but differ in topology and gate order. Our AO21 uses an AOI21 stage followed by an inverter and keeps both diffusion rows continuous. For MUX2, the stage break and the space needed to cross the gate band account for the extra site. Other layouts also need wider gaps than HD where a routing channel carries the output across the `npc` band.

The resistance column in Table 3 sums the extracted resistors assigned to the output net. Parallel branches affect that sum, so it cannot be read as the effective resistance from the drains to the pin. The capacitance column includes the capacitors incident on the output, including coupling capacitors. Across the 33 generated layouts, the plans record 599 licon cuts at 353 diffusion contact locations. In the 190-contact sample used to evaluate contact arrays, 70 locations kept a single licon because nearby wires left insufficient space for a column. Installed HD typically uses two cuts in the NMOS row and three in the PMOS row. The cuts help reduce contact resistance where there is room for them, while the larger landings can add capacitance.

Adding a parallel path raised the summed output-net resistance by 18% in `a22oi_1` and 17% in `o22ai_1`, even though the pin-to-drain resistance did not increase. We use the characterized timing to assess the effect of routing changes on delay.

4.4 Nominal delay and slew

For the nine detailed comparisons, Table 4 uses the HD table point nearest 100 ps input slew and 10 fF output load. The input slew is about 122.5 ps for each cell, and the exact load depends on the installed cell's table indices. We interpolate the generated tables onto that point and take the maximum delay and output slew over the applicable arcs and edge directions. The final column gives the median signed timing-table difference reported by the matched extraction and characterization flow. It describes a different statistic from the nominal worst arc.

Table 4: Nominal timing of the nine generated reference layouts compared with installed HD layouts characterized by the same flow.

Cell	Load (fF)	Area change (%)	Delay (ps)	Slew (ps)	Median diff. (%)
inv_1	9.52	0.0	123/123	94/94	−0.75
nand2_1	9.13	0.0	139/139	105/106	−0.40
nor2_1	6.56	0.0	177/174	150/147	+0.19
nand3_1	8.56	0.0	154/156	121/123	−1.19
nor3_1	11.09	0.0	369/373	379/385	−2.90
a21oi_1	6.09	0.0	212/213	176/178	−1.96
o21ai_1	6.35	+25.0	208/207	175/175	−0.78
a21o_1	9.42	0.0	239/228	104/94	+5.32
mux2_1	9.30	+11.1	289/369	262/106	−3.16

At the nominal point, AOI21 reaches the installed width with nearly the same worst delay and slew. The single-stage medians are within 2.91% of the recharacterized HD tables. MUX2 shows why delay and slew are both worth reporting. Its worst delay is 80 ps shorter at the nominal point, while its worst output transition takes 262 ps against 106 ps for HD. Our MUX2 drives its output directly from a complex gate, whereas the installed cell has an output inverter. That difference is consistent with the slower transition, although the extracted comparison also includes changes in device arrangement and wiring.

Table 5 gives the HD comparison for the other six functions with installed equivalents. The function-matched benchmark uses each HD cell's own nominal point. The table also gives the realized widths and the timing of the layouts in the 33-cell set at the common generated-cell point of 122.5 ps and 9.13 fF. The two sets of measurements use different layouts or loads, so the benchmark columns are the basis for the reported HD timing comparisons. For AOI22, OAI22, and XOR3, extraction of the installed cell stopped at a pin label. The shipped Liberty values are available for all six.

Table 5: Additional function-matched HD benchmarks and generated results. Benchmark delay and slew are generated / shipped HD / recharacterized HD, in ps. A dash indicates unavailable extraction.

	Function-matched benchmark		Generated layouts in the 33-cell set		
Cell	Delay	Slew	Sites / HD	Area change	Delay / slew (ps)
a22oi_1	273/258/—	238/223/—	5/6	−16.7%	292/265
o22ai_1	384/336/—	390/341/—	5/5	0.0%	272/249
and2_1	158/214/203	107/102/108	5/5	0.0%	155/100
maj3_1	299/412/394	108/105/110	12/8	+50.0%	293/96
xor2_1	250/220/218	218/177/182	6/7	−14.3%	287/269
xor3_1	403/462/—	247/116/—	14/19	−26.3%	371/228

4.5 Characterizer calibration

Table 6 summarizes the comparison with shipped Liberty for six installed cells. Both Yosys and OpenSTA loaded all six Liberty files produced by the fixed build. Its median relative error met the 5% tolerance, and the largest error in input capacitance met the 10% tolerance. The full-ramp control increased the median timing error to 7.46%, which the comparison rejected.

The largest percentage error in both builds occurs on the AOI21 B1 falling arc at 1.5 ns input slew and 0.5 fF output load. The shipped delay is 3.25 ps, while the fixed and stock builds give 14.34 and 7.03 ps respectively. At this slow-slew, light-load point, the two 50% crossings are close together, so a difference of 11.09 ps gives a large relative error. The fixed build improves the median across the tables even though its largest relative error is higher.

The stock BSIM4 code overwrites the stress-adjusted low-field mobility while applying the

Table 6: Calibration against the shipped Liberty tables.

Metric	Fixed ngspice 47	Stock ngspice 42
Delay and transition entries compared	2,744	2,744
Median relative timing error	3.36%	4.83%
Maximum relative timing error	341%	116%
Maximum input-capacitance error	5.46%	5.61%

instance mobility multiplier. The patch applies the multiplier to the value that already includes the stress adjustment. In the design-level calibration, replacing only the six installed Liberty entries changed the serial-parallel multiplier's worst path delay from 0.6883 to 0.6890 ns. The same change was observed when the original mapped netlist was held fixed and retimed with the replacement entries.

4.6 Mapping and gate-level simulation

Both ABC scripts produced equivalent netlists for all 33 cells. At least one script selected the generated cell in 20 cases. That includes the five requests for which the reduced HD library had no single-cell equivalent (`f14_1`, `f15_1`, `maj4_1`, `maj4_opt_1`, and `maj4n_1`), as well as 15 of the other 28 requests. The per-cell results and summary are in `results/round_trip/`.

The supplied `scripts/map.sh` can exit successfully without mapping the design under Yosys 0.52, which splits its inline ABC script at semicolons. Our round-trip script reads the mapping commands and passes each ABC script as a separate file.

The design-level benchmark in Table 7 asks whether the mapper uses the new cells when they are offered alongside the full shipped library. This experiment used a 20-cell generated library. The two mapped serial-parallel multiplier netlists were run through the same self-checking Icarus testbench with functional cell models. OpenSTA timed both designs at a 1.25 ns clock period. Substituting a generated XOR2 model with one wrong truth-table row made the gate-level simulation fail.

Table 7: Serial-parallel multiplier benchmark with the shipped and augmented libraries.

Metric	HD only	HD plus generated cells
Mapped instances	327	296
Generated instances	0	160
Mapped cell area (μm^2)	2,945.3	2,790.2
Gate-level functional simulation	Pass	Pass
Worst path delay (ns)	0.6883	0.7755

In this benchmark the mapper reduced area by 5.3%, while the worst path delay increased by 12.7%. A generated XOR2 appears on that path. The comparison is at the mapped-netlist level, so the reported area is the sum of library cell areas and the timing comes from static timing analysis of the mapped design.

5 Discussion

5.1 Choosing a NOR3 size

NOR3 provides a direct sizing comparison because its rising transition must charge the output through three series PMOS devices. The folded layout triples the width of each series PMOS and splits it into three fingers. The NMOS widths are the same in both layouts, and the measurements in Table 8 use the same 122.5 ps input slew and 9.13 fF output load.

The folded cell reduces rising delay by 32% to 42%, depending on the input. Falling delay increases by 14% to 25%, even though the NMOS widths are unchanged. The wider PMOS devices

Table 8: Extracted NOR3 timing and input load under the two sizing policies.

Metric	`hd`	`stack`
Width (sites)	4	10
Area (μm^2)	5.00	12.51
Total NMOS / PMOS width (μm)	1.95/3.00	1.95/9.00
Rising delay A / B / C (ps)	330/319/284	223/208/165
Falling delay A / B / C (ps)	100/94/90	124/114/102
Rising output slew (ps)	326	175–182
Input capacitance across pins (fF)	2.16–2.29	4.45–4.65

and the additional layout capacitance still load the output during a falling transition. We would use the `hd` version as the default NOR3 because it occupies less area and places a smaller load on the preceding gate. The folded version makes sense as a separate drive option for a path limited by a rising transition. The measured differences include both sizing and the geometry needed to implement it.

An optional `folded_finger_nets_on_met1` setting routes the source and drain nets of folded fingers on first metal. For `nor3_stack_1`, it reduced the extracted signal-net resistance sum by 6.4% and increased signal-net capacitance by 4.1%. We leave the setting disabled because it would route signals on a layer the handout reserves for rails in single-stage cells.

The frame uses high-threshold PMOS devices. The isolated rising-delay cost of that device choice remains unmeasured. Quantifying it requires two extractions or matched simulation decks with the same sizing, parasitics, input slew, and load, changing only the PMOS threshold-voltage model.

5.2 Internal-stage sizing

The extracted results give little support for enabling the Elmore-based width search by default. At the common nominal point, optimized MAJ4 improves from 351 to 332 ps. AO21, MUX2, and XOR2 change by at most 1.0%, while MAJ3 and AND2 become 19.8% and 34.4% slower, respectively. All six optimized layouts retain the site count of their `hd` counterpart. We excluded `xor3_opt_1` because layout planning took too long. The model uses a resistance and capacitance estimate for each stage but does not propagate the input slope from one stage to the next. Increasing a device width can therefore look favorable in the estimate while the larger load and the actual waveform at the internal node offset that gain in the extracted circuit.

5.3 Output disturbances and charge sharing

Among the 612 directed transitions for which the output was expected to stay at the same logic value, the largest excursion was 448 mV in `mux2_opt_1`. Both data inputs were low, and the select input fell from high to low. The output briefly rose before returning low and stayed below the 0.90 V half-supply threshold, so the cell passed without a circuit change. The saved output extrema establish the size of the disturbance. Distinguishing charge sharing from capacitive coupling or a temporary conduction path would require inspecting the internal node waveforms for that edge.

6 Limitations

6.1 Synthesis, routing, and pin access

The synthesizer builds repeated sub-stages separately, which can leave duplicate logic in a generated cell.

For multi-stage cells, the planner searches the available `li1`-only placements and widths before allowing signal `met1`. The search includes rip-up and reroute when enabled. The base `xor2_1`, `xor3_1`, `maj3_1`, and `maj4_1` layouts still need first metal for signals. These layouts have no assigned `hd_reference` in the detailed comparison, so its wiring criterion does not apply to them even where

there is an installed cell with the same function. The single-stage `maj4n_1` also uses signal first metal, which the handout reserves for rails in single-stage cells. The plan flags that exception as `met1_in_single_stage_cell`.

The negotiated router works on `li1`. It cannot resolve a crossing forced by the selected gate order by adding width alone. In the selected XOR2 order, for example, the connection from output Y between the rows must cross the connection between the two A gates. Failed local-interconnect attempts add planning time. Turning off `rip_up_with_negotiated` and enabling `lone_net_check` made planning about two to three times faster for cells that use first metal in the planning benchmark. MAJ3 still took about four minutes, and XOR3 took well over ten.

First-metal pin access can add width beyond that needed for the routing channels. Adjacent input pins can be 0.48 μm apart, while a first-metal wire needs a 0.42 μm gap between two access pads. Preserving access leaves MAJ3 at 12 sites against eight for HD and `maj4n_1` at ten sites. We check the local access pads. Access in a fully placed and routed design remains untested.

6.2 Verification and runtime

The built-in DRC control places a narrow first-metal stripe at a fixed location in the first abutment context. The full request set detects the stripe, but the control can miss it if the first generated cell already has first metal at that location. A refused request returns status `fail` with the reader's diagnostic. The response was based on the available passing example, and its conformity to the full Appendix A failure format remains unverified.

Parallel characterization is limited by memory. With eight concurrent ngspice processes, `make char` peaked at 6.9 GB out of the 7.7 GB available to WSL. Systems with less memory can lower `INNER_JOBS` or `JOBS` in `local.mk`. A Windows file lock reported as error 13 or 22 triggers a retry of the affected step. The NAND2 view-generation step needed one such retry and passed on its second attempt.

6.3 Coverage of the library and design benchmarks

The electrical checks cover one process, voltage, and temperature corner. They do not cover process variation, noise margins across corners, antenna effects, electromigration, or ESD. The extracted functional tests exercise one input transition at a time, and the multiplier testbench checks logic behavior with zero-delay models. The 3.36% calibration median describes typical agreement across six reference tables and does not bound every timing entry.

The augmented-library design test in Step 9 covers the 20-cell multiplier benchmark. The full 33-cell library has per-cell mapping results, with the design-level gate simulation and OpenSTA comparison still outstanding for that library. The detailed HD comparison also excludes `a22oi_1`, `o22ai_1`, `and2_1`, `maj3_1`, and `xor2_1` because they lack an assigned `hd_reference`. Their timing comparisons in the report come from the separate function-matched benchmark.

In the earlier workflow, `make all` stopped at model verification, and characterization, final views, the round trip, and the negative tests were run separately with `make char views roundtrip test_views`. The current `make all` includes characterization, final views, and the per-cell round trip. The negative tests are included in `make test`.

7 Contributions

- **Youssef Ayman Hassan** developed Boolean minimization, multi-stage CMOS synthesis, and transistor sizing, including the Elmore-based search for internal-stage widths. He implemented GDS generation and worked on layout optimizations such as parallel contact cuts and contact-to-via alignment.

- **Hassan Ashraf** developed transistor ordering with paired Euler trails, stage placement, and interconnect routing. He worked on movable-contact routing with negotiated congestion,

improved layout-planning runtime, and verified layouts with DRC, LVS, and comparisons against installed HD cells.

- **Yara Abdelkader** developed the request reader, characterization and calibration against the HD library, and the functional test flow. She also worked on batching transistor-level simulations, parallelizing verification, exporting Liberty and JSON responses, checking the ABC mapping round trip, and making the flow and demo portable.

Youssef and Hassan worked together on folded gate placement and first-metal pin access. Youssef and Yara ran the 33-cell verification and characterization flow and documented the results.

A Toolchain and result files

Table 9: Tool versions recorded in the project sources and outputs.

Tool	Version
Python	3.12
GDSfactory / KLayout / Matplotlib	9.51.0 / 0.30.12 / 3.11.2
Magic / Netgen	8.3.684 / 1.5.133
ngspice, characterization	47 with the BSIM4 stress-mobility fix
ngspice, stock calibration comparison	42
Icarus Verilog / Yosys / OpenSTA	12.0 / 0.52 / 2.0.17
ABC	Yosys toolchain build, standalone revision unrecorded
PDK	SKY130A with `sky130_fd_sc_hd`

Per-cell layout plans, characterized tables, and response files are under `cells/`. The physical-verification summaries are in `results/drc/verification.json` and `results/lvs/verification.json`, while the transistor-level verdicts and waveforms' measured extrema are under `results/transistor_level/`. The sizing comparison in Table 8 is in `results/sizing/nor3.json`.

References

[1] R. L. Maziasz and J. P. Hayes. "Exact width and height minimization of CMOS cells." *Proceedings of the 28th ACM/IEEE Design Automation Conference*, pp. 487–493, 1991.

[2] L. McMurchie and C. Ebeling. "PathFinder: A negotiation-based performance-driven router for FPGAs." *Proceedings of the ACM International Symposium on Field-Programmable Gate Arrays*, 1995.